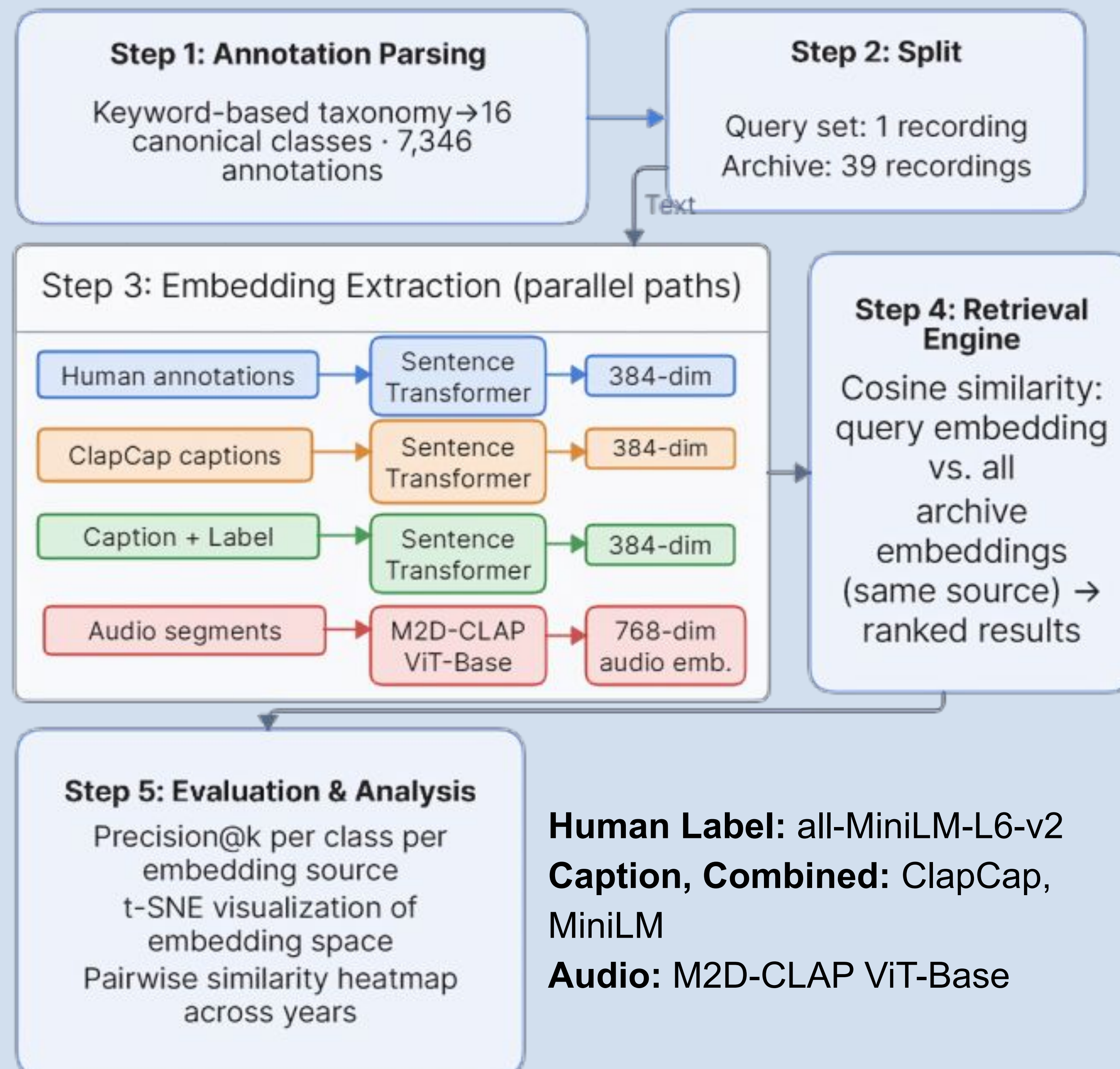


Problem Statement

Passive acoustic monitoring generates vast environmental recordings that are **difficult to search manually**. Existing pretrained audio and language models struggle with out-of-domain field recordings due to domain mismatch and variability in annotations. This study compares four audio embedding strategies for semantic similarity search using 40 two-hour recordings from Vogelinsel Nature Reserve, Germany (2017–2024).

Methods



Precision@k measures the fraction of top-k retrieved annotations sharing the same canonical class as the query.

Results

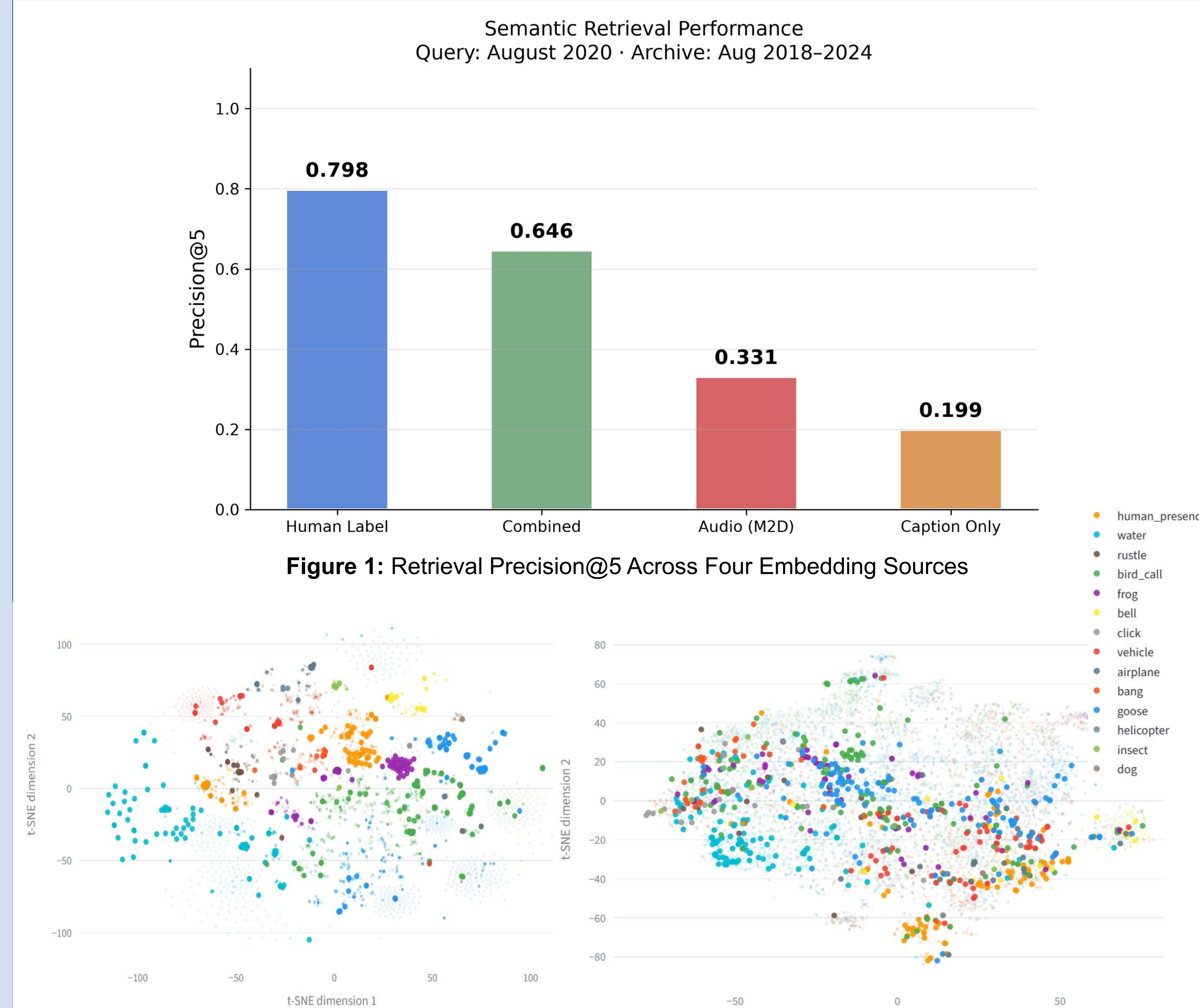


Figure 2: t-SNE visualization of embedding space colored by sound class. Human label embeddings (left) show distinct class clusters, while M2D audio embeddings (right) show poor class separation (consistent with retrieval results).

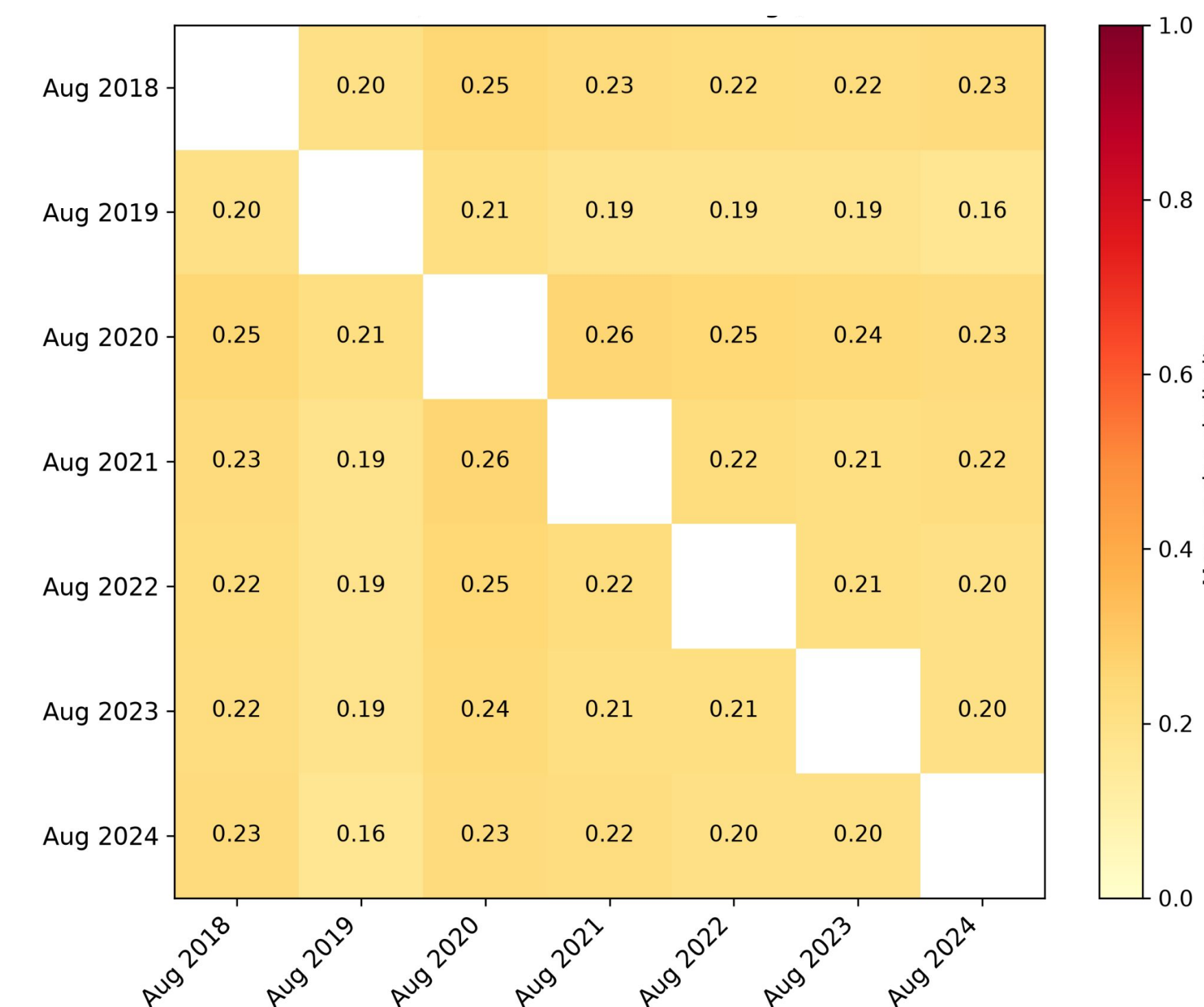


Figure 3: Pairwise cosine similarity between August recordings (2018–2024) using human label embeddings.

Conclusions

- Human label embeddings achieve Precision@5 of 0.798, significantly outperforming ClapCap captions (0.199) and M2D audio features (0.331)
- ClapCap systematically generates generic outdoor descriptions ("birds chirping in the background"), collapsing semantically distinct classes into overlapping embedding regions
- M2D audio embeddings show domain mismatch, showing high precision on distinctive classes but weak overall
- Text-based embeddings encode seasonal acoustic patterns, suggesting human annotations carry ecological signal beyond class labels

Future Work

- Understanding the impact of shallow fine-tuning on domain-specific annotations and whether it reduces the gap between pretrained model accuracy and their zero-shot performance
- Spatial audio analysis using DSP methods to capture directional and perceptual audio characteristics for improved environmental sound retrieval

